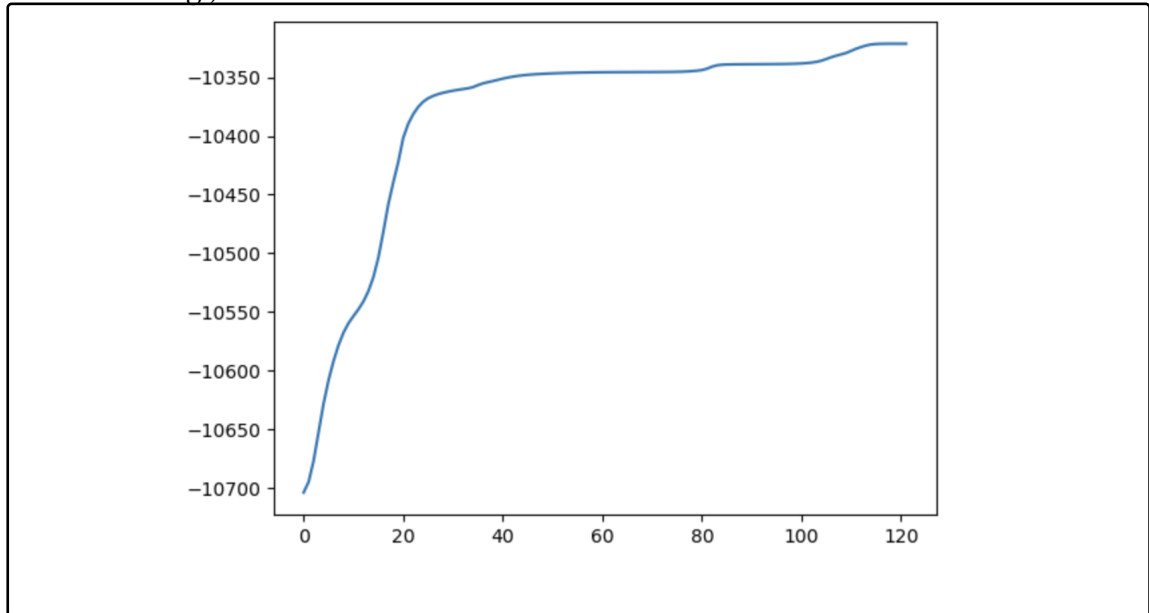


6. [Implement the EM algorithm for Gaussian mixture models 30 pts]. Apply this to the expression data for mouse HIP brain tissue from Homework 2. Use only the first 10 mice (the first 10 rows in the data matrix) and cluster the genes.

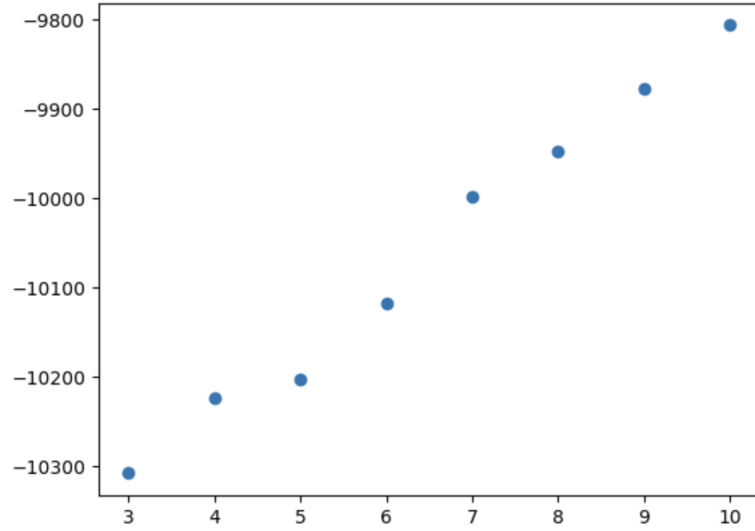
- (a) (10 pts) Assume  $K = 3$  clusters. Use the initialization provided in Homework 2 to initialize the means (use the first 10 rows for 10 mice). Use a  $10 \times 10$  diagonal matrix with 1.0 along the diagonals to initialize all covariance matrices. Use  $[0.3, 0.3, 0.4]$  to initialize the mixing proportions. Plot the log-likelihood of data  $\log p(\text{Data}) = \sum_{i=1}^N \log p(x_i)$ , where  $N$  is the number of genes, over iterations. (Hint: The data log-likelihood should always go up over iterations. If this values goes down even slightly, this means your code has a bug!)



- (b) (10 pts) Using the model you estimated in (a) above, compute the probability of the first gene to belong to each of the three clusters. (There should be three values in the answer).

3.78467096e-05, 1.22716580e-01, 8.77245574e-01

- (c) (10 pts) Run the EM algorithm, assuming  $K = 3, \dots, 10$  clusters. Plot the log-likelihood of the data across different values for  $K$ . What do you think is the best choice for the number of clusters?



I think the best choice of clusters is the one with the highest log likelihood value. In this case, it is 10 clusters.